

Lecture 4: Frequentist mean analysis

Bayesian & Frequentist Probability

ODSL / LMU

Known covariance; confidence from repeated experiments

Lectures 4–5 summary

1. Model and sampling distribution

Observe n independent d -component vectors with an unknown mean and a **known**, positive definite covariance:

$$X_i \stackrel{\text{iid}}{\sim} \mathcal{N}_d(\mu, \Sigma), \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

The mean μ is fixed; the observations and any estimator computed from them vary across repeated experiments. Lower-case x_i and $\bar{x}$ denote the values in the observed dataset.

The maximum-likelihood estimator is the sample mean:

$$\hat{\mu} = \bar{X}, \quad \hat{\mu} \mid \mu \sim \mathcal{N}_d(\mu, \Sigma/n).$$

It is unbiased, $\mathbb{E}_\mu[\hat{\mu}] = \mu$, and its covariance is Σ/n . Individual observations scatter with covariance Σ ; estimates from repeated datasets scatter with covariance Σ/n . Standard errors decrease as $n^{-1/2}$.

2. Likelihood and precision

For fixed observed data, the likelihood is a function of the candidate mean:

$$L(\mu) \propto \exp\left[-\frac{n}{2}(\bar{x} - \mu)^\top \Sigma^{-1}(\bar{x} - \mu)\right].$$

Define

$$\chi^2(\mu) = \sum_{i=1}^n (x_i - \mu)^\top \Sigma^{-1}(x_i - \mu).$$

Subtracting its minimum gives the likelihood-ratio statistic:

$$\begin{aligned} \Delta\chi^2(\mu) &= \chi^2(\mu) - \chi^2(\bar{x}) \\ &= -2 \log \frac{L(\mu)}{L(\bar{x})} \\ &= n(\bar{x} - \mu)^\top \Sigma^{-1}(\bar{x} - \mu). \end{aligned}$$

The Fisher information is $F = n\Sigma^{-1}$. The sample mean attains the Cramér–Rao bound, $\text{Cov}(\hat{\mu}) = F^{-1}$: it has minimum covariance among unbiased estimators in this model.

3. Mahalanobis distance and chi-square

For a single $X \sim \mathcal{N}_d(\mu, \Sigma)$, whitening gives $Z = \Sigma^{-1/2}(X - \mu) \sim \mathcal{N}_d(0, I)$. Hence

$$Q = (X - \mu)^\top \Sigma^{-1}(X - \mu) = \sum_{j=1}^d Z_j^2 \sim \chi_d^2.$$

This is a **squared Mahalanobis distance**: it accounts for both unequal variances and correlations. For the sample mean, use Σ/n instead. Evaluated at the true mean $\mu_\star$,

$$\Delta\chi^2(\mu_\star) \sim \chi_d^2.$$

The degrees of freedom here are the number d of fitted mean components. This statement is exact for the assumed Gaussian model, at any n ; it is not a large-sample approximation.

4. Constructing a confidence region

Choose a confidence level $q = 1 - \alpha$ and the chi-square quantile $\chi_{d,q}^2$. Retain the candidate means satisfying

$$C_q(x) = \{\mu : \Delta\chi^2(\mu) \leq \chi_{d,q}^2\}.$$

The observed region is centred on $\bar{x}$; it can be computed without knowing the true mean. In d dimensions it is an ellipsoid, oriented by Σ and shrinking as $n^{-1/2}$. If λ_j are the eigenvalues of Σ , its semiaxis lengths are $\sqrt{\chi_{d,q}^2 \lambda_j / n}$.

For $d = 1$, with known variance σ^2 , the interval is

$$\bar{x} \pm z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}, \quad 95\%: \bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}},$$

where z_p is the standard-normal p quantile.

For 95% regions, $\chi_{1,0.95}^2 = 3.841$ and $\chi_{2,0.95}^2 = 5.991$. Thus the 2D joint region has Mahalanobis radius $\sqrt{5.991} \simeq 2.448$; radius 2 contains only 86.5% in 2D. Probability content depends on dimension.

A 95% interval for a single component is $\bar{x}_j \pm 1.96\sqrt{\Sigma_{jj}/n}$. These marginal intervals are different from a joint 95% ellipsoid.

5. What the confidence level means

Across independent repetitions at any fixed true mean,

$$\mathbb{P}_{X|\mu_\star}\{\mu_\star \in C_q(X)\} = q.$$

The **region varies; the true mean stays fixed**. A 95% procedure includes that mean in 95% of repeated experiments in the long run. A finite simulation will fluctuate around 95%.

After observing the data, the particular region either contains $\mu_\star$ or does not. The confidence level describes the procedure's coverage; it does not assign a posterior probability to the parameter in this observed region.

To check coverage, keep $\mu_\star$ and Σ fixed, simulate many fresh datasets of size n , recompute each region, and count the fraction containing $\mu_\star$. Evaluate $\Delta\chi^2$ at $\mu_\star$ when checking its reference distribution.

6. Assumptions and unknown variance

Exact coverage above uses independent Gaussian observations and the correct, known Σ . A misspecified noise model can invalidate the claimed coverage.

Separate case: unknown scalar variance. For iid Gaussian data with $n > 1$, let $s^2 = \sum_i (x_i - \bar{x})^2 / (n - 1)$. Then

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}, \quad C_q(x) = \bar{x} \pm t_{n-1, 1-\alpha/2} \frac{s}{\sqrt{n}},$$

where S denotes the random sample standard deviation and s its observed value. This is an exact frequentist interval despite the nuisance parameter σ^2 . Frequentist methods can handle nuisance parameters without assigning them a prior; Bayesian marginalisation is a different treatment of that uncertainty.

Lecture 5: Bayesian mean analysis

Known covariance; learning about a mean

Lectures 4–5 summary

1. Model, likelihood and prior

Observe n independent d -component vectors with an unknown mean and a **known**, positive definite covariance:

$$x_i \stackrel{\text{iid}}{\sim} \mathcal{N}_d(\mu, \Sigma), \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

The sample mean contains all the information about μ in this model, and its sampling distribution is

$$\bar{X} \mid \mu \sim \mathcal{N}_d(\mu, \Sigma/n).$$

For fixed observed data, the likelihood is

$$L(\mu) \propto \exp \left[-\frac{n}{2} (\bar{x} - \mu)^\top \Sigma^{-1} (\bar{x} - \mu) \right].$$

Bayes' theorem combines this with a prior:

$$p(\mu \mid x_{1:n}) \propto L(\mu) p(\mu).$$

The likelihood varies the candidate mean while holding the data fixed; the posterior is a normalised density over that mean.

2. Gaussian prior: an analytic posterior

Take the conjugate prior $\mu \sim \mathcal{N}_d(m_0, C_0)$, where C_0 is positive definite. Completing the square gives

$$\mu \mid x_{1:n} \sim \mathcal{N}_d(m_n, C_n),$$

$$C_n^{-1} = C_0^{-1} + n\Sigma^{-1},$$

$$m_n = C_n (C_0^{-1} m_0 + n\Sigma^{-1} \bar{x}).$$

Posterior precision = prior precision + data precision. The posterior mean is a precision-weighted combination of the prior mean and the sample mean. In this Gaussian case it also equals the maximum-a-posteriori estimate:

$$\hat{\mu}_{\text{MAP}} = \arg \max_{\mu} p(\mu \mid x_{1:n}) = m_n.$$

3. How the prior changes the answer

A tighter prior pulls the posterior towards m_0 and reduces its covariance. Shifting m_0 with C_0 fixed moves the posterior mean without changing C_n .

In the broad-prior limit $C_0^{-1} \rightarrow 0$,

$$m_n \rightarrow \bar{x}, \quad C_n \rightarrow \Sigma/n.$$

The Bayesian and frequentist Gaussian regions then have the same boundaries but different interpretations. A narrower credible region need not have better coverage at a particular fixed truth.

4. Credible regions and their dimension

A region $B_q(x)$ is credible at level q when

$$\int_{B_q(x)} p(\mu \mid x) d\mu = q.$$

A **highest-posterior-density (HPD) region** selects the largest posterior density values until their total mass is q . For the Gaussian posterior, the joint HPD ellipsoid is

$$(\mu - m_n)^\top C_n^{-1} (\mu - m_n) \leq \chi_{d,q}^2,$$

where $\chi_{d,q}^2$ is the q quantile with d degrees of freedom. For a joint 95% region, the cutoff is 3.841 in one dimension and 5.991 in two dimensions. A unit Mahalanobis radius contains only about 39.3% in two dimensions, not 68.3%. The marginal posterior for component j is $\mathcal{N}(m_{n,j}, C_{n,jj})$, giving a 95% marginal interval

$$m_{n,j} \pm 1.96 \sqrt{C_{n,jj}}.$$

Marginal intervals and a joint ellipsoid are different regions; their probability levels are not interchangeable.

5. Posterior mass versus coverage

Fixed truth: hold μ_* fixed and redraw the data. Coverage is $\mathbb{P}_{X \mid \mu_*} \{\mu_* \in B_q(X)\}$; it need not equal q . Prior shrinkage can shift the posterior centre systematically relative to this truth.

Prior-predictive experiment: draw $\mu \sim p(\mu)$, then $X \sim p(X \mid \mu)$, using the same prior and likelihood as the analysis. The probability of covering each dataset's own truth is

$$\begin{aligned} \mathbb{P}\{\mu \in B_q(X)\} &= \mathbb{E}_X [\mathbb{P}\{\mu \in B_q(X) \mid X\}] \\ &= q. \end{aligned}$$

Calibration averaged over the matching prior does not imply nominal coverage separately at every fixed mean. This is a check within the assumed model, not evidence that the model describes real data correctly.

6. From a mean to physical parameters

If the mean is predicted by $\mu = \mu(\theta)$, then

$$p(\theta \mid x) \propto p(\theta) \exp \left[-\frac{n}{2} r(\theta)^\top \Sigma^{-1} r(\theta) \right], \quad r(\theta) = \bar{x} - \mu(\theta).$$

A nonlinear mean generally gives a non-Gaussian posterior, even with Gaussian noise. Numerical integration or sampling may be needed. Likelihood and prior assumptions matter; collecting more data does not automatically fix a misspecified model. Lecture 6 relaxes the assumption that the covariance is known.